MATH 8090:
Univariate Volatility Modeling

Reading: *An Introduction to Analysis of Financial Data with R* (2013), Ruey S. Tsay

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Week 10

Agenda

Background

ARCH Model

GARCH Model

IGARCH and EGARCH Models

Stochastic Volatility Model

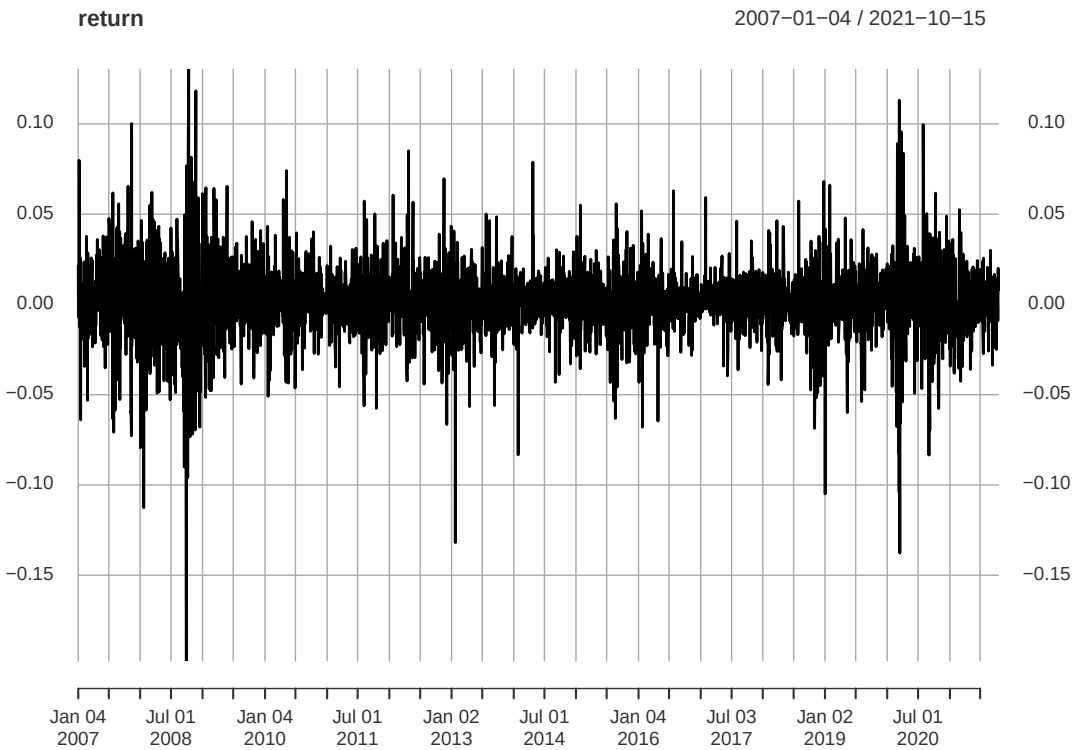
Financial Time Series: Stock Prices



How can we characterize and predict the fluctuations in stock prices over time?

From Stock Prices to Log Returns

$$r_t = \log\left(\frac{y_t}{y_{t-1}}\right), \quad \text{where } y_t \text{ is the price at time } t$$



Volatility Clustering: Large fluctuations tend to occur together, followed by periods of relatively small fluctuations.

From ARMA Models to Volatility Models

Recall: ARMA models describe dependence in the conditional mean.

$$r_t = \mu_t + a_t, \qquad \mathbb{E}(a_t \mid \mathcal{F}_{t-1}) = 0,$$

where $\mathcal{F}_{t-1}$ denotes all information available up to time $t-1$.

Financial returns often exhibit:

- Little autocorrelation in returns.
- Serial dependence in squared returns.
- Volatility clustering: large and small fluctuations tend to occur in clusters.

Central Question

Can we predict the **magnitude** of future fluctuations even when their direction is difficult to predict?

Modeling Volatility

Volatility is the conditional standard deviation of returns, measuring their variability given past information.

Why is volatility important?

- Option pricing - e.g., Black-Scholes model
- Risk management - e.g., Value-at-Risk (VaR)
- Asset allocation - e.g., Minimum-variance portfolios
- Forecasting - e.g., Prediction intervals for returns

Key challenge: Volatility is unobserved and changes over time.

Modeling Conditional Volatility

Recall: Decompose returns into a conditional mean and an innovation:

$$r_t = \mu_t + a_t,$$

where

$$\mu_t = \mathbb{E}(r_t|\mathcal{F}_{t-1})$$

Our focus: Model the conditional variance of returns:

$$\mathrm{Var}(r_t|\mathcal{F}_{t-1}) = \mathrm{Var}(a_t|\mathcal{F}_{t-1}) = \sigma_t^2.$$

- **ARMA models**: Describe dependence in the conditional mean.
- **Volatility models**: Describe how conditional variance evolves over time.

Literature on (Univariate) Volatility Modeling

ARCH/GARCH family:

- Autoregressive conditional heteroscedastic (ARCH) model [Engle, 1982]: Variance depends on past squared shocks
- Generalized Autoregressive Conditional Heteroskedasticity (GARCH) Model [Bollerslev, 1986]: Adds dependence on past conditional variance
- IGARCH: Models highly persistent volatility with a unit root in the variance dynamics
- EGARCH [Nelson, 1991]: Allows asymmetric responses to positive and negative shocks

Alternative framework:

- Stochastic Volatility[1] (SV): Volatility follows a latent stochastic process

[1] [Melino and Turnbull, 1990; Harvey, Ruiz, and Shephard, 1994; Jacquier, Polson, and Rossi, 1994]

ARCH(m) Model

An ARCH(m) model specifies the conditional variance using past squared innovations

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \cdots + \alpha_m a_{t-m}^2,$$

where:

- $\{\epsilon_t\}$ are i.i.d. with $\mathbb{E}[\epsilon_t] = 0$ and $\mathrm{Var}[\epsilon_t] = 1$
- $\alpha_0 > 0$ and $\alpha_i \geq 0$ for $1 \leq i \leq m$

AR Analogy: ARCH(m) implies an AR(m)-type representation for the squared innovations.

ARCH(1) Model

Consider the simplest ARCH model:

$$a_t = \sigma_t \epsilon_t$$
$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2$$

Interpretation:

- α_0 determines the baseline conditional variance
- α_1 measures the effect of the previous squared shock on current volatility
- A large absolute shock at time $t-1$ leads to higher conditional variance at time t

Key observation: Positive and negative shocks of the same magnitude have identical effects on conditional variance

Properties of ARCH(1)[2]

Conditional moments:

- $\mathbb{E}(a_t|\mathcal{F}_{t-1}) = 0$
- $\mathrm{Var}(a_t|\mathcal{F}_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2$

Unconditional moments:

- $\mathbb{E}(a_t) = 0$
- $\mathrm{Var}(a_t) = \dfrac{\alpha_0}{1-\alpha_1}$

Implications:

- Innovations are uncorrelated but have time-varying conditional variance
- Squared innovations may be serially correlated
- ARCH(1) exhibits heavy tails under Gaussian innovations (when the fourth moment exists)

[2] Assume $0 \leq \alpha_1 < 1$

Building an ARCH Model

1. **Model the mean and test for ARCH effects**

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   - Fit a mean model and obtain residuals: $a_t = y_t - \hat{\mu}_t$
   - Test for ARCH effects using the Ljung–Box test on $\{a_t^2\}$ or Engle's Lagrange Multiplier test

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3. **Estimate model parameters**: Use Conditional Maximum Likelihood Estimation

4. **Check model adequacy**
   - Examine standardized residuals and their squares
   - Check for remaining ARCH effects and assess the innovation distribution

Advantages and Limitations of ARCH Models

Advantages:

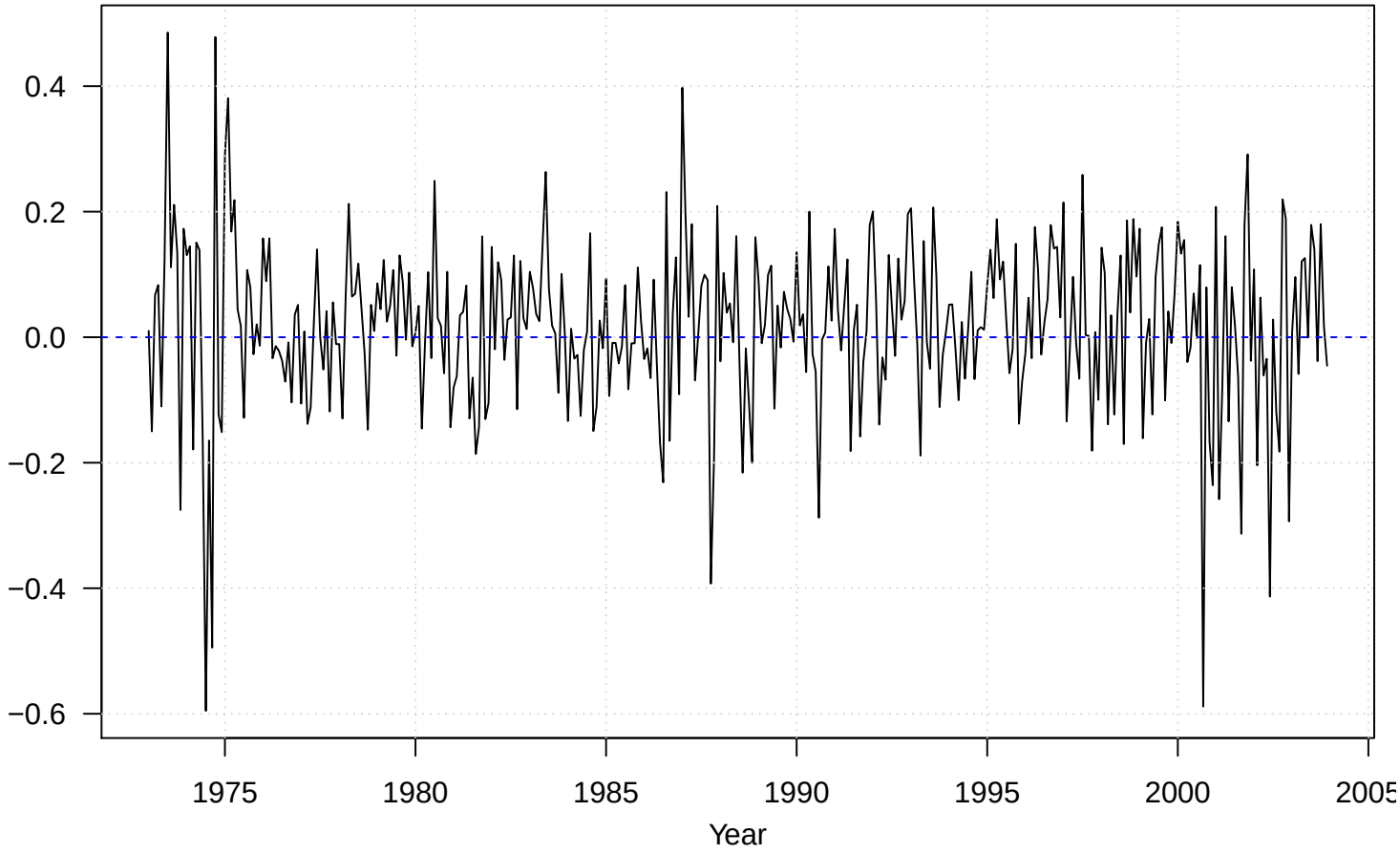
- **Simple and intuitive:** Easy to interpret and estimate
- **Captures volatility clustering:** Models periods of high and low volatility
- **Generates heavy tails:** Helps explain extreme returns

Limitations:

- **Symmetric response:** Positive and negative shocks have equal effects
- **Restrictive moment conditions:** For ARCH(1) with Gaussian innovations, a finite fourth moment requires $3\alpha_1^2 < 1$
- **Limited persistence:** Capturing persistent volatility may require a high ARCH order

Example: Monthly Log Returns of Intel Stock

We use monthly log returns of Intel to illustrate ARCH modeling



Question: Do periods of large and small returns tend to cluster together?

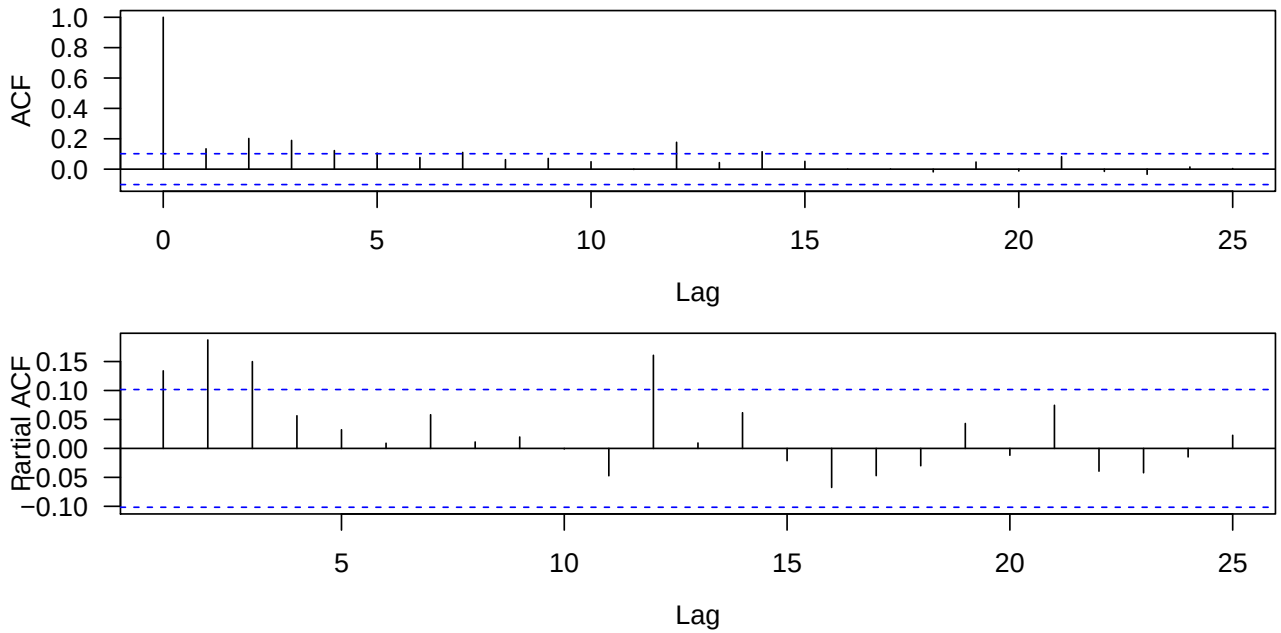
Detecting ARCH Effects

We examine squared residuals to assess dependence in volatility

```
> Box.test(y^2, lag = 12, type = 'Ljung')

        Box-Ljung test

data:  y^2
X-squared = 68.67, df = 12, p-value = 5.676e-10
```



Key questions:

- Is there evidence of ARCH effects?
- What ARCH order might be appropriate?

Fitting an ARCH(3) Model

We first fit an ARCH(3) model with Gaussian innovations:

$$r_t = \mu + a_t, \quad a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{3} \alpha_i a_{t-i}^2,$$

assuming $\epsilon_t \overset{i.i.d.}{\sim} \mathrm{N}(0,1)$.

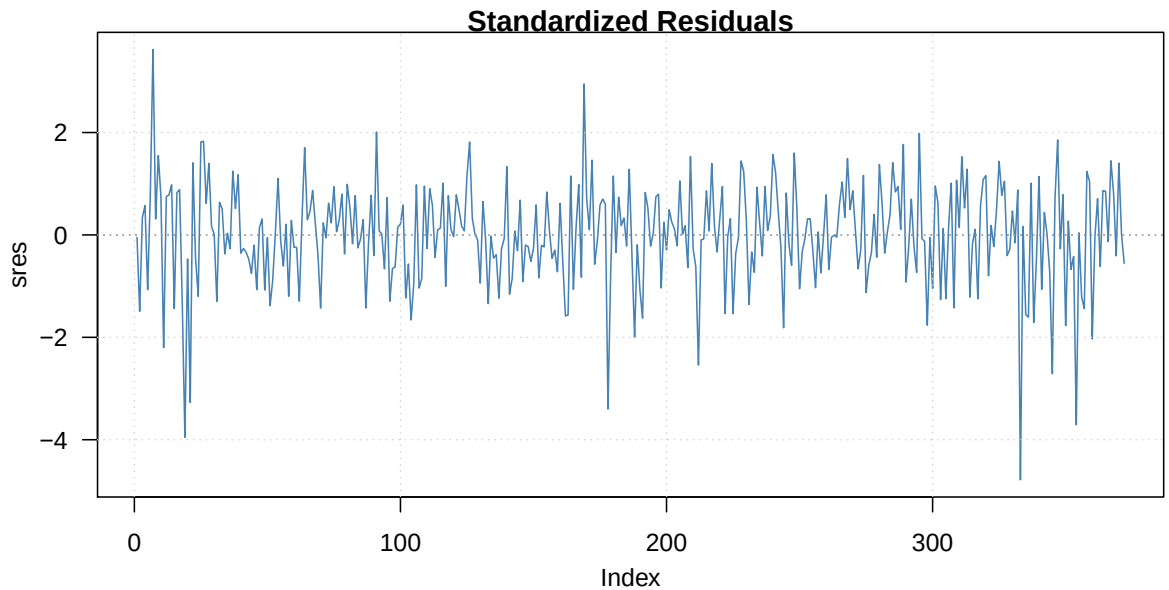
```
Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.016572    0.006423    2.580  0.00988 **
omega   0.012043    0.001579    7.627  2.4e-14 ***
alpha1  0.208649    0.129177    1.615  0.10626
alpha2  0.071837    0.048551    1.480  0.13897
alpha3  0.049045    0.048847    1.004  0.31536
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Question: Are all three ARCH terms necessary?

Simplifying to ARCH(1)

We fit a more parsimonious ARCH(1) model for comparison

```
Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.016570    0.006161    2.689  0.00716 **
omega   0.012490    0.001549    8.061 6.66e-16 ***
alpha1  0.363447    0.131598    2.762  0.00575 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```



Questions:

- Does ARCH(1) adequately capture volatility clustering?
- Do the standardized residuals show any remaining patterns?

Checking the ARCH(1) Model

We assess the adequacy of the fitted ARCH(1) model using standardized residuals

Standardised Residuals Tests:

			Statistic	p-Value
Jarque-Bera Test	R	Chi^2	122.404	0
Shapiro-Wilk Test	R	W	0.9647625	8.273101e-08
Ljung-Box Test	R	Q(10)	13.72604	0.1858587
Ljung-Box Test	R	Q(15)	22.31714	0.09975386
Ljung-Box Test	R	Q(20)	23.88257	0.2475594
Ljung-Box Test	R^2	Q(10)	12.50025	0.25297
Ljung-Box Test	R^2	Q(15)	30.11276	0.01152131
Ljung-Box Test	R^2	Q(20)	31.46404	0.04935483
LM Arch Test	R	TR^2	22.036	0.0371183

Diagnostic tests:

- Jarque–Bera & Shapiro-Wilk Tests: Assess normality
- Lagrange multiplier (LM) Test: Detect remaining ARCH effects

ARCH(1) with Student-t Innovations

Financial returns often exhibit heavy tails, motivating an ARCH(1) model with Student-t innovations

```
Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.021571    0.006054    3.563 0.000366 ***
omega   0.013424    0.001968    6.820 9.09e-12 ***
alpha1  0.259867    0.119901    2.167 0.030209 *
shape   5.985979    1.660030    3.606 0.000311 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Log Likelihood:
 242.9678    normalized:  0.6531391

Description:
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Standardised Residuals Tests:
                             Statistic p-Value
 Jarque-Bera Test   R    Chi^2  130.8931  0
 Shapiro-Wilk Test  R    W      0.9637533 5.744995e-08
 Ljung-Box Test     R    Q(10)  14.31288  0.1591926
 Ljung-Box Test     R    Q(15)  23.34043  0.07717449
 Ljung-Box Test     R    Q(20)  24.87286  0.2063387
 Ljung-Box Test     R^2  Q(10)  15.35917  0.1195054
 Ljung-Box Test     R^2  Q(15)  33.96318  0.003446127
 Ljung-Box Test     R^2  Q(20)  35.46828  0.01774746
 LM Arch Test       R    TR^2   24.11517  0.01961957
```

GARCH Model

Motivation: GARCH extends ARCH by incorporating past conditional variances, allowing more persistent volatility dynamics

Model:

$$a_t = \sigma_t \epsilon_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{i=1}^{m} \alpha_i a_{t-i}^2 + \sum_{j=1}^{s} \beta_j \sigma_{t-j}^2.$$

Assumptions:

- ϵ_t are i.i.d. with mean 0 and variance 1
- $\alpha_0 > 0$, $\alpha_i, \beta_j \geq 0$
- $\sum_{i=1}^{\max(m,s)} (\alpha_i + \beta_i) < 1$ ensures a finite unconditional variance

Interpretation:

- **Past squared shocks**: Capture the impact of recent innovations
- **Past conditional variances**: Capture volatility persistence

Key takeaway: GARCH can capture persistent volatility with fewer parameters than a high-order ARCH model

GARCH(1,1): Properties and Persistence

Model:

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

Key Properties:

- **Finite unconditional variance:** Requires $\alpha_1 + \beta_1 < 1$, with nonnegative coefficients
- **Volatility clustering:** Large shocks increase future conditional variance
- **Volatility persistence:** $\alpha_1 + \beta_1$ governs the persistence of volatility shocks
- **Heavy tails:** Gaussian GARCH innovations can produce unconditional excess kurtosis

One-step-ahead variance forecast:

$$\sigma_t^2(1) = \alpha_0 + \alpha_1 a_t^2 + \beta_1 \sigma_t^2$$

GARCH(1,1): Multi-Step Variance Forecasts

Define the persistence parameter: $\rho = \alpha_1 + \beta_1$

One-step forecast:

$$\begin{aligned}\sigma_t^2(1) &= \alpha_0 + \alpha_1 a_t^2 + \beta_1 \sigma_t^2 \\ &= \alpha_0 + \underbrace{(\alpha_1 + \beta_1)}_{\rho} \sigma_t^2 + \alpha_1 \sigma_t^2 (\epsilon_t^2 - 1)\end{aligned}$$

Multi-step recursion:

$$\sigma_t^2(\ell) = \alpha_0 + \rho \sigma_t(\ell - 1)^2, \quad \ell \geq 2$$

Closed-form forecast:

$$\sigma^t(\ell) = \bar{\sigma}^2 + \rho^{\ell - 1} \left[\sigma_t^2(1) - \bar{\sigma}^2 \right],$$

where $\bar{\sigma}^2 = \frac{\alpha_0}{1-\rho}$ is the unconditional variance

Key Takeaway: Variance forecasts converge to the unconditional variance, with the rate of convergence governed by ρ

Intel Example Revisited - GARCH(1,1) Fitting

We revisit Intel's monthly log returns and fit a GARCH(1,1) model

```
Error Analysis:
        Estimate  Std. Error  t value Pr(>|t|)
mu     0.0163276   0.0062624    2.607  0.00913 **
omega  0.0010918   0.0005291    2.063  0.03907 *
alpha1 0.0802716   0.0281162    2.855  0.00430 **
beta1  0.8553014   0.0461374   18.538  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

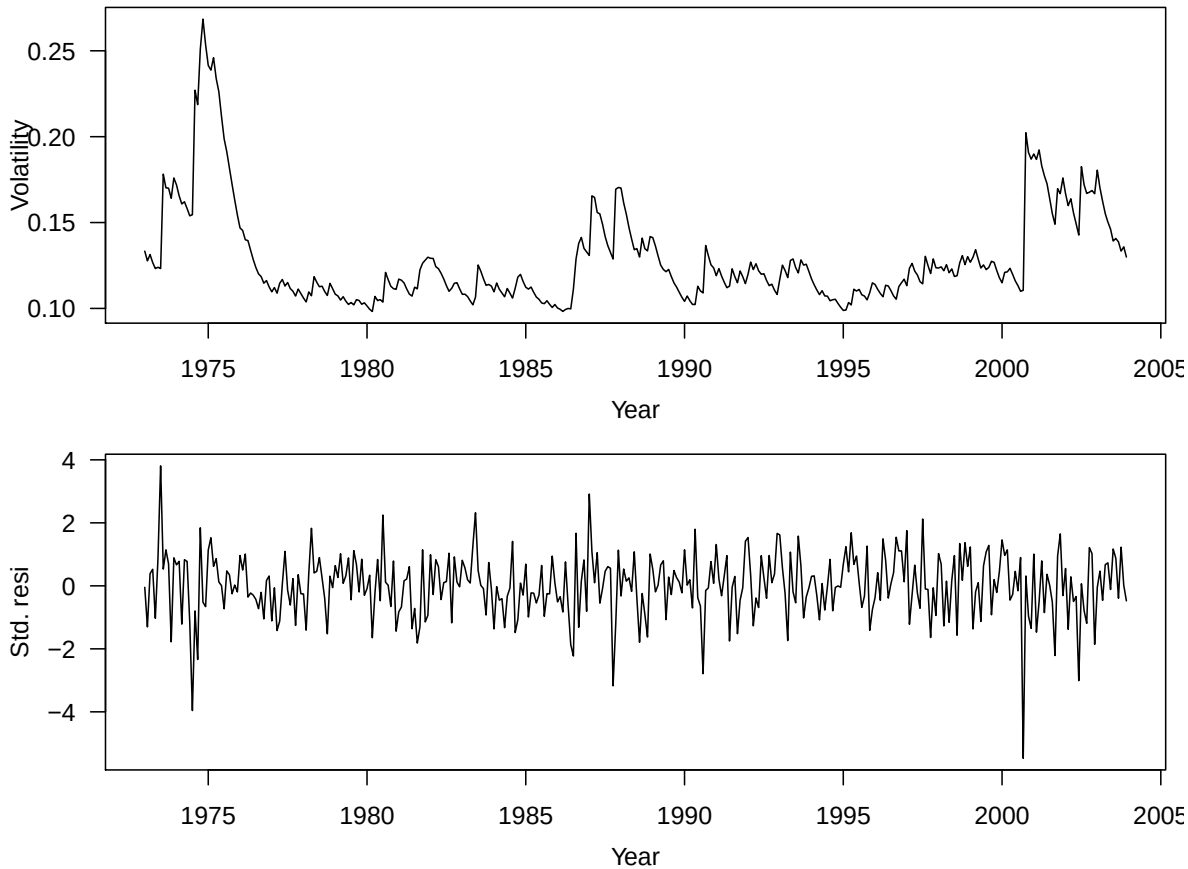
Log Likelihood:
 239.5189    normalized:  0.6438681

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Standardised Residuals Tests:
                             Statistic p-Value
 Jarque-Bera Test   R    Chi^2  156.5138  0
 Shapiro-Wilk Test  R    W      0.9676933 2.471139e-07
 Ljung-Box Test     R    Q(10)  9.805485  0.4577215
 Ljung-Box Test     R    Q(15)  16.54435  0.346824
 Ljung-Box Test     R    Q(20)  17.8005   0.6005484
 Ljung-Box Test     R^2  Q(10)  0.5130171 0.9999925
 Ljung-Box Test     R^2  Q(15)  10.24557  0.8040151
 Ljung-Box Test     R^2  Q(20)  11.77988  0.9234441
 LM Arch Test       R    TR^2   9.334459  0.6741288
```

How persistent is volatility, as measured by $\rho = \alpha_1 + \beta_1$?

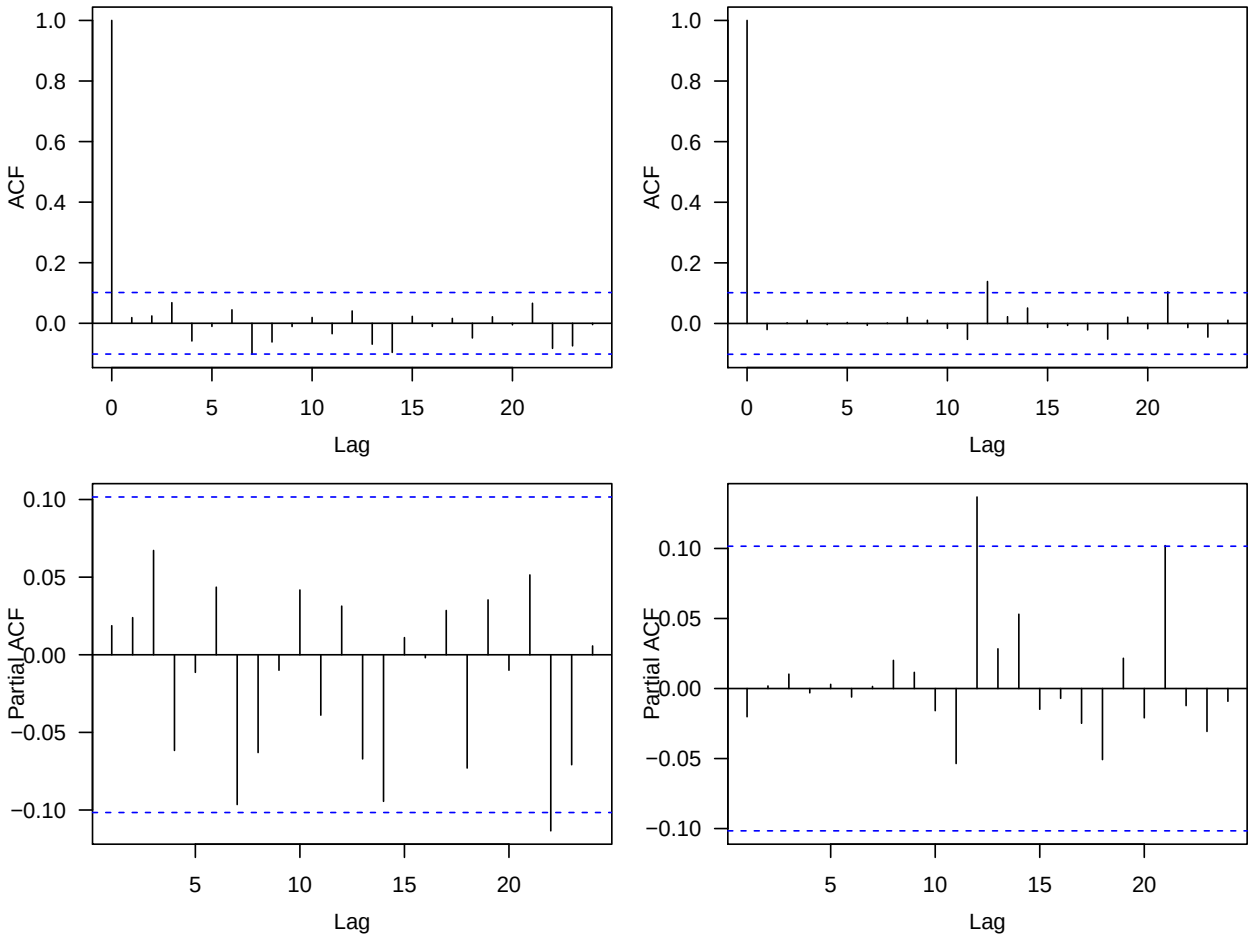
Estimated Volatility and Standardized Residuals



Interpretation:

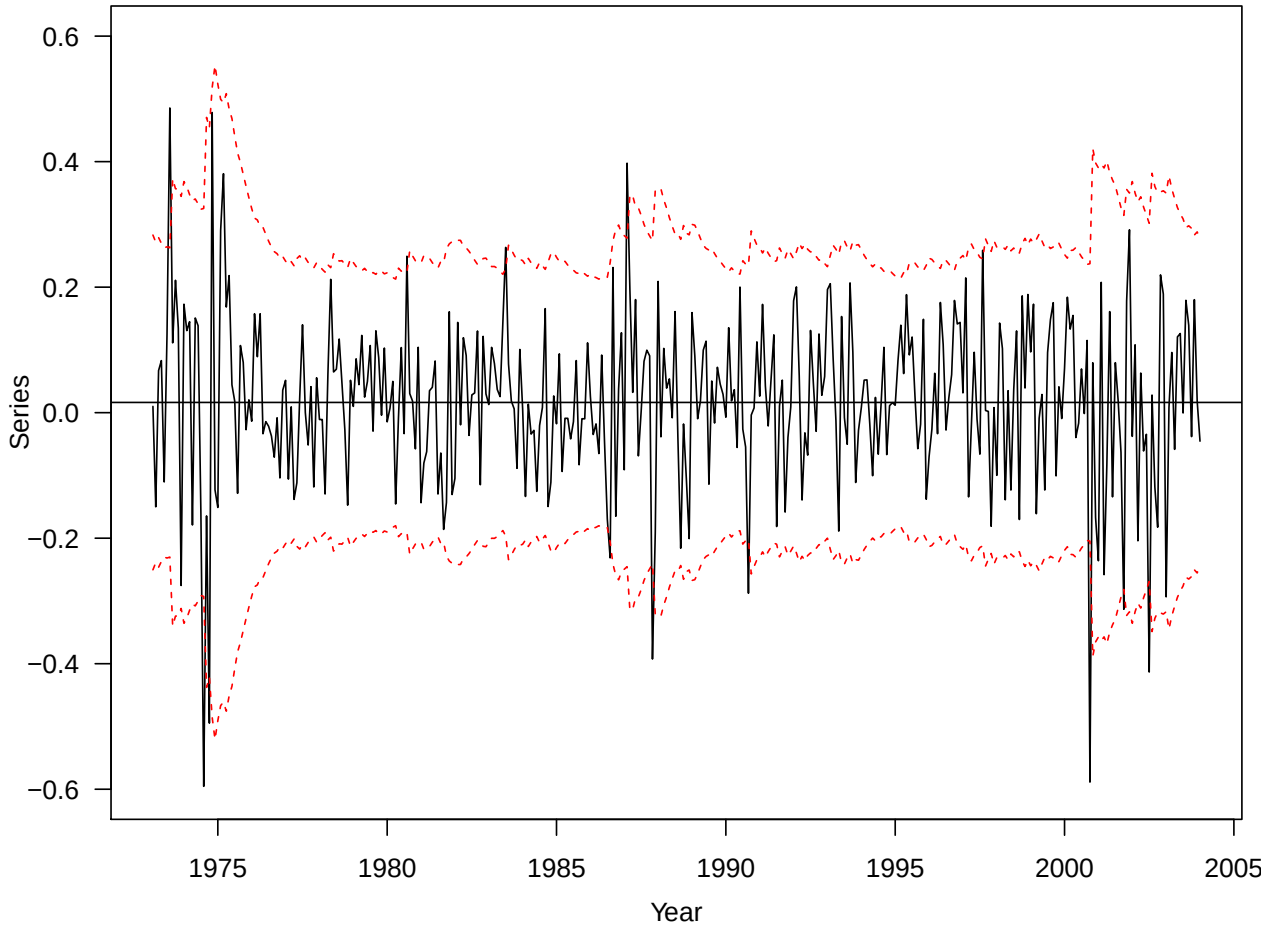
- **Conditional volatility**: Highlights periods of high and low variability
- **Standardized residuals**: Should show no remaining dependence or volatility clustering

GARCH(1,1) Model Diagnostics



A well-fitted GARCH model should leave little serial correlation in both standardized and squared standardized residuals

Time-Varying Prediction Intervals



Interpretation:

- Prediction intervals widen with higher volatility and narrow with lower volatility
- Connection to VaR: The lower predictive tail measures risk

Integrated GARCH (IGARCH)

Key idea: Volatility shocks have persistent effects when the GARCH coefficients sum to one (unit root)

IGARCH(1, 1) model:

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t^2 = \alpha_0 + \beta_1 \sigma_{t-1}^2 + (1 - \beta_1) a_{t-1}^2$$

Volatility forecasts:

$$\sigma_t^2(\ell) = \sigma_t(1)^2 + (\ell - 1)\alpha_0, \quad \ell \geq 1$$

Interpretation:

- Forecasts increase linearly with the horizon when $\alpha_0 > 0$
- Unlike stationary GARCH, volatility forecasts do not revert to a finite long-run variance

Why Exponential GARCH (EGARCH)?

Limitation of standard GARCH:
Positive and negative shocks of equal magnitude have the same effect on conditional volatility

Financial market behavior:
Negative returns often increase future volatility more than equally large positive returns.

EGARCH [Nelson, 1991] addresses this through:

- Asymmetry: Different volatility responses to positive and negative shocks
- Log-variance modeling: Ensures positive conditional variance without nonnegative coefficient constraints

Key question: Does the sign of a return shock matter for future volatility?

EGARCH: Modeling Asymmetric Shocks

Innovation function:

$$g(\epsilon_t) = \theta\epsilon_t + \gamma\big(|\epsilon_t| - \mathbb{E}|\epsilon_t|\big), \quad \text{where } \mathbb{E}[g(\epsilon_t)] = 0$$

Interpretation:

- If $\gamma > 0$, negative returns ($\epsilon_t < 0$) cause higher volatility than positive ones.
- The term $|\epsilon_t| - \mathbb{E}|\epsilon_t|$ ensures mean zero.

Asymmetric form:

$$g(\epsilon_t) = \begin{cases} (\theta + \gamma)\epsilon_t - \gamma\mathbb{E}|\epsilon_t|, & \epsilon_t \geq 0, \\ (\theta - \gamma)\epsilon_t - \gamma\mathbb{E}|\epsilon_t|, & \epsilon_t < 0 \end{cases}$$

EGARCH(1, 1) Model

Conditional log-variance:

$$a_t = \sigma_t \epsilon_t, \text{ where } \epsilon_t \text{ are } i.i.d. N(0,1)$$

$$\log(\sigma_t^2) = (1-\alpha)\alpha_0 + \alpha \log(\sigma_{t-1}^2) + g(\epsilon_t),$$

where $g(\epsilon_t) = \theta\epsilon_t + \gamma(|\epsilon_t| - \mathbb{E}|\epsilon_t|)$

Transforming back to the variance scale

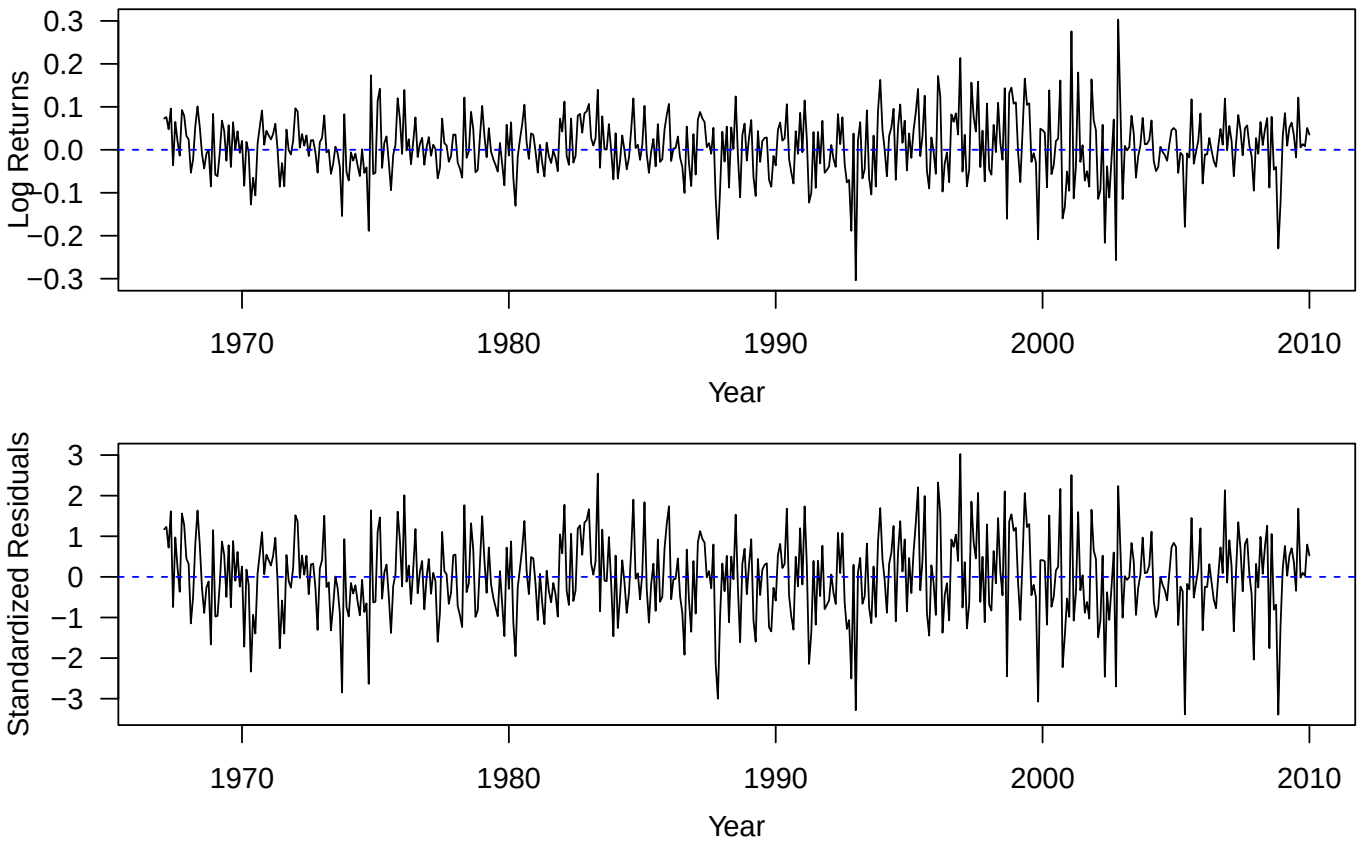
$$\sigma_t^2 = \sigma_{t-1}^{2\alpha} \exp\left((1-\alpha)\alpha_0 - \sqrt{\frac{2}{\pi}}\gamma\right) \begin{cases} \exp\left[(\gamma+\theta)\epsilon_{t-1}\right] & \text{if } \epsilon_{t-1} \geq 0, \\ \exp\left[(\gamma-\theta)\epsilon_{t-1}\right] & \text{if } \epsilon_{t-1} < 0. \end{cases}$$

Interpretation:

- α: Volatility persistence
- $\gamma + \theta$: Response to positive shocks
- $\gamma - theta$: Response to negative shocks

Example: IBM Stock Returns

Data: Monthly log returns of IBM stock, January 1967December 2009



Question: Do negative returns have a stronger impact on future volatility than positive returns?

EGARCH(1,1) Fit: IBM Returns

Fitted mean model:

$$r_t = 0.067 + a_t,$$

where $a_t = \sigma_t \epsilon_t$ and $\epsilon_t \sim \mathrm{N}(0, 1)$.

Fitted volatility model:

$$\log(\sigma_t^2) = -0.598 + 0.920 \log(\sigma_{t-1}^2) + 0.218(|\epsilon_{t-1}| - 0.423\epsilon_{t-1})$$

Therefore, we have

$$\sigma_t^2 = \sigma_{t-1}^{2\times 0.920} \exp(-0.598) \times \begin{cases} \exp(0.125\epsilon_{t-1}) & \text{if } \epsilon_{t-1} \geq 0, \\ \exp(-0.310\epsilon_{t-1}) & \text{if } \epsilon_{t-1} < 0. \end{cases}$$

EGARCH: Quantifying Asymmetry

Compare two standardized shocks: 2 and $+2$

$$\frac{\sigma_t^2(\epsilon_{t-1} = -2)}{\sigma_t^2(\epsilon_{t-1} = 2)} = \frac{\exp\left(-0.31 \times (-2)\right)}{\exp\left(0.125 \times 2\right)} = e^{0.37} = 1.448$$

Interpretation:
A negative shock of two standard deviations produces approximately 44.8% higher conditional variance than an equally large positive shock

Takeaway: EGARCH captures the stronger volatility response to negative market shocks

Stochastic Volatility (SV) Model

A simple stochastic volatility (SV) model assumes time-varying volatility driven by a latent process:

$$a_t = \sigma_t \epsilon_t, \qquad (1 - \alpha_1 B - \cdots - \alpha_m B^m) \log(\sigma_t^2) = \alpha_0 + \nu_t$$

where:

- $\epsilon_t \sim$ i.i.d. $N(0, 1)$ are shocks to returns
- $\nu_t \sim$ i.i.d. $N(0, \sigma_\nu^2)$ are shocks to volatility
- ϵ_t and ν_t are independent

Key idea: Volatility follows its own stochastic process rather than being a deterministic function of past shocks (as in GARCH)

From GARCH to Long-Memory Stochastic Volatility

ARCH/GARCH models use past shocks and variances to model volatility, while stochastic volatility models treat volatility as a latent stochastic process

Long-Memory Stochastic Volatility (LMSV) Model

$$a_t = \sigma_t \epsilon_t, \quad \sigma_t = \sigma \exp(u_t/2)$$
$$(1-B)^d u_t = \eta_t, \quad 0 < d < 0.5,$$

where:

- ϵ_t are independent standard normal innovations
- η_t are independent normal innovations with variance σ_η^2, independent of ϵ_t
- u_t is a latent long-memory process governing log-volatility

Key feature: Unlike standard ARCH/GARCH models, LMSV allows volatility dependence to decay slowly over time

LMSV: Log Transformation and Inference

Taking logarithms of squared innovations:

$$\begin{aligned}\log(a_t^2) &= \log(\sigma^2) + u_t + \log(\epsilon_t^2) \\ &= \underbrace{\left[\log(\sigma^2) + \mathbb{E}(\log(\epsilon_t^2))\right]}_{\mu} + u_t + \underbrace{\left[\log(\epsilon_t^2) - \mathbb{E}(\log(\epsilon_t^2))\right]}_{e_t}\end{aligned}$$

Interpretation:

- u_t: Gaussian long-memory signal
- e_t: Independent, non-Gaussian observation noise

Thus, log-squared innovations follow a latent long-memory signal-plus-noise model

Inference challenge: The volatility process u_t is unobserved, requiring integration over latent states for likelihood-based inference.

- Bayesian inference: MCMC
- Likelihood-based alternatives: numerical approximations and specialized filtering methods